

Answer 1: ^{2/3} Because we can show every thing with
Sets. language

Answer 2, $E = \{x: x \in \mathbb{Z}\}$, $E = \{2^n: n \in \mathbb{Z}\}$

Answer 3: ① = $E = \{2^n: n \in \mathbb{N}\}$

② = $E = \{2^n: n \in \mathbb{Z}\}$?

③ = $E = \{x: 3 \leq x \leq 8, x \in \mathbb{N}\}$

Answer 4: ① = $|\{x \in \mathbb{Z}: |x| < 10\}| = 19$

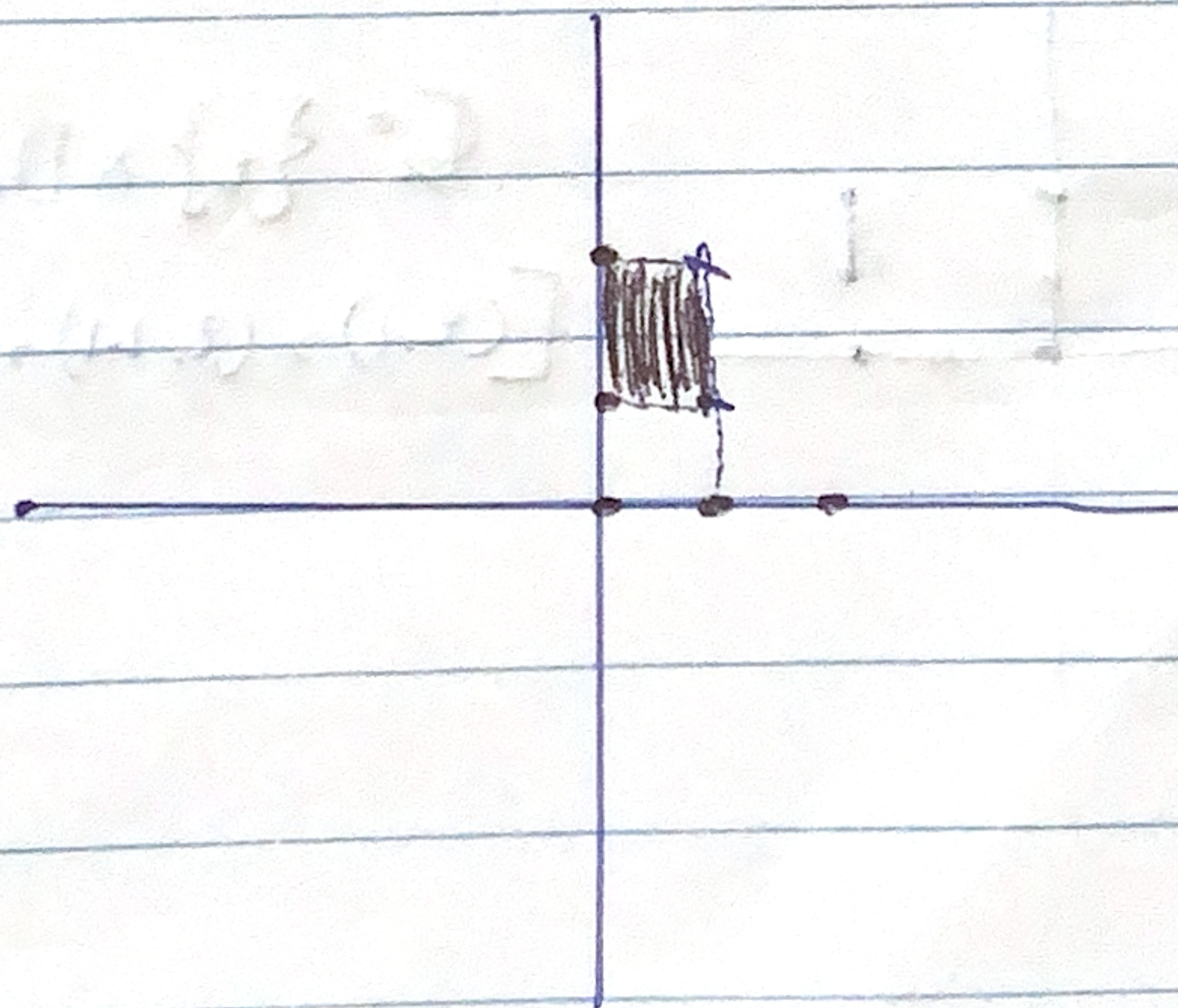
② = $|\{x \in \mathbb{R}: |x| < 10\}| =$

$\{0, 0.1, 0.2, \dots, 8.9999\} = \infty$

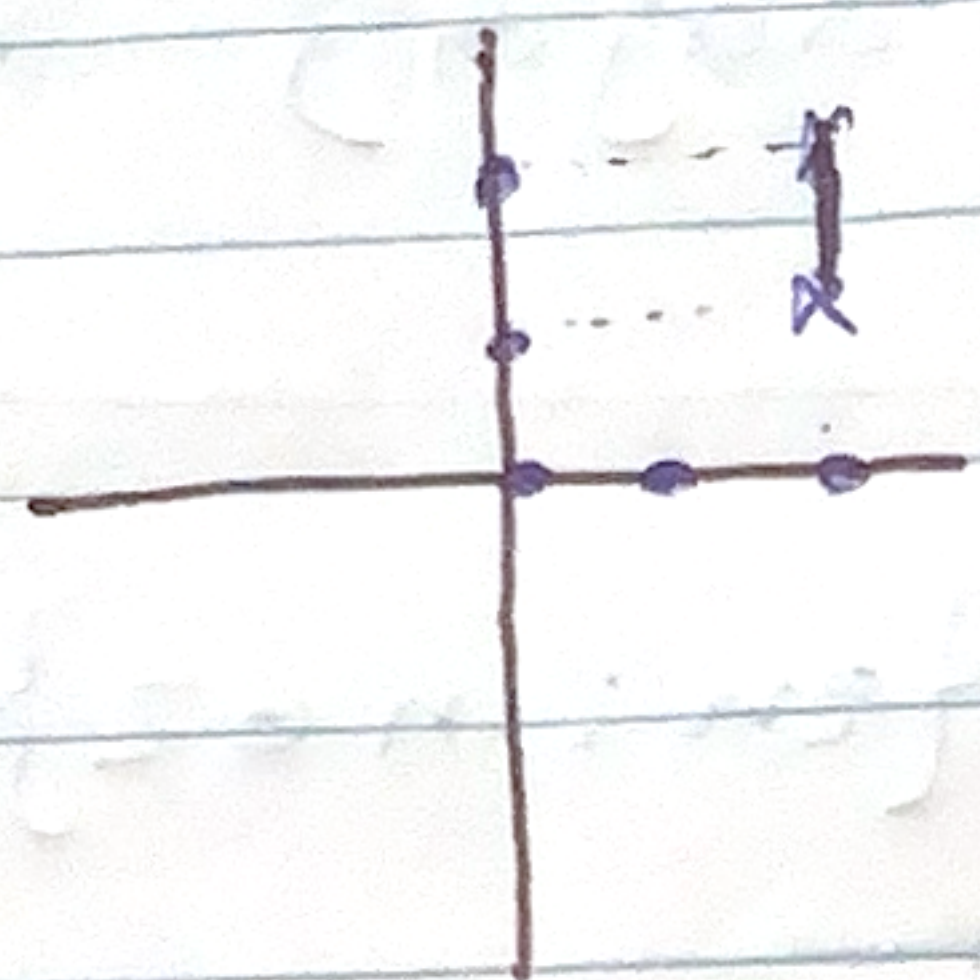
③ = $|\{x \in \mathbb{N}: 5x \leq 20\}| =$

$|\{1, 2, 3, 4\}| = 4$

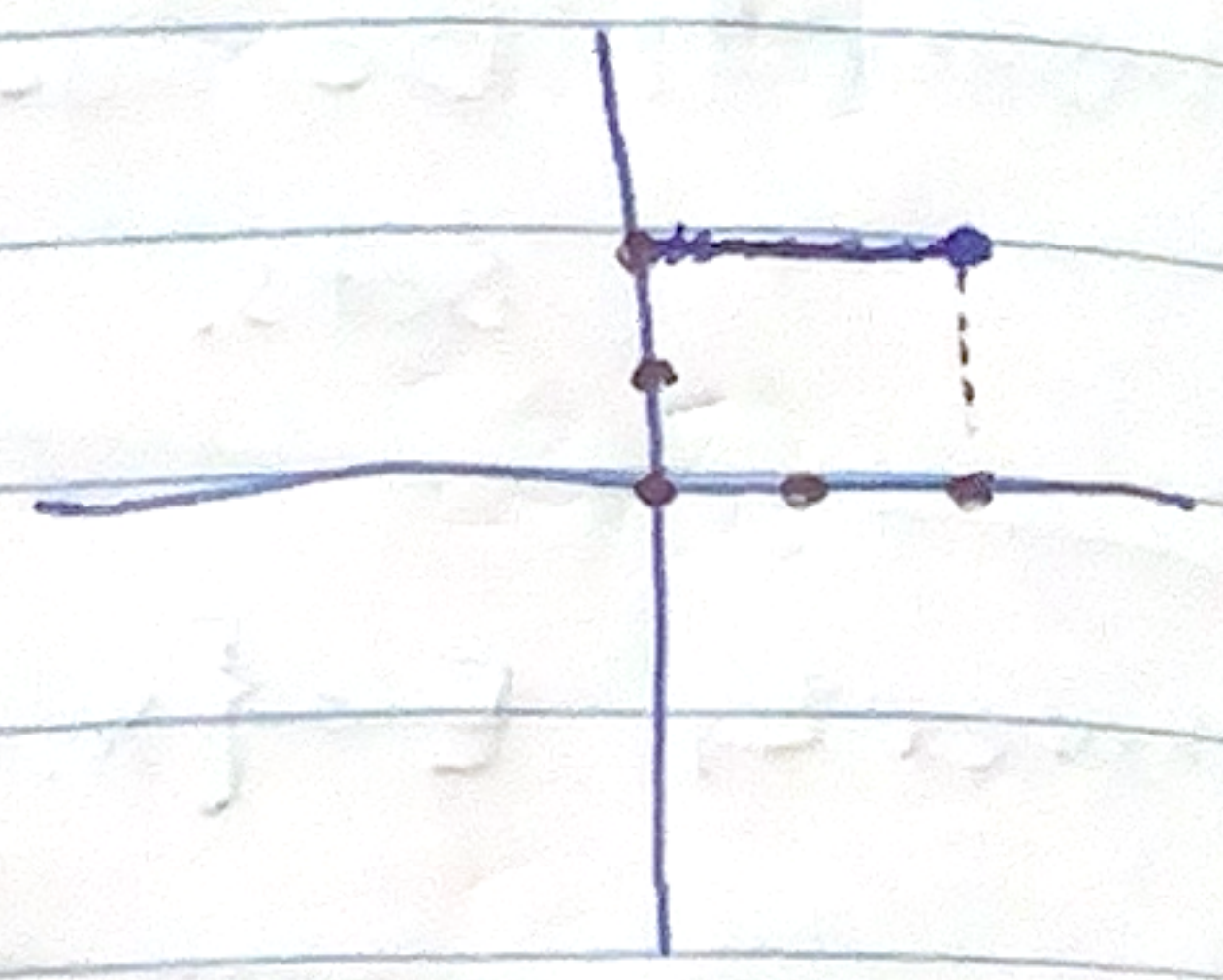
Answer 5: ①



(1)



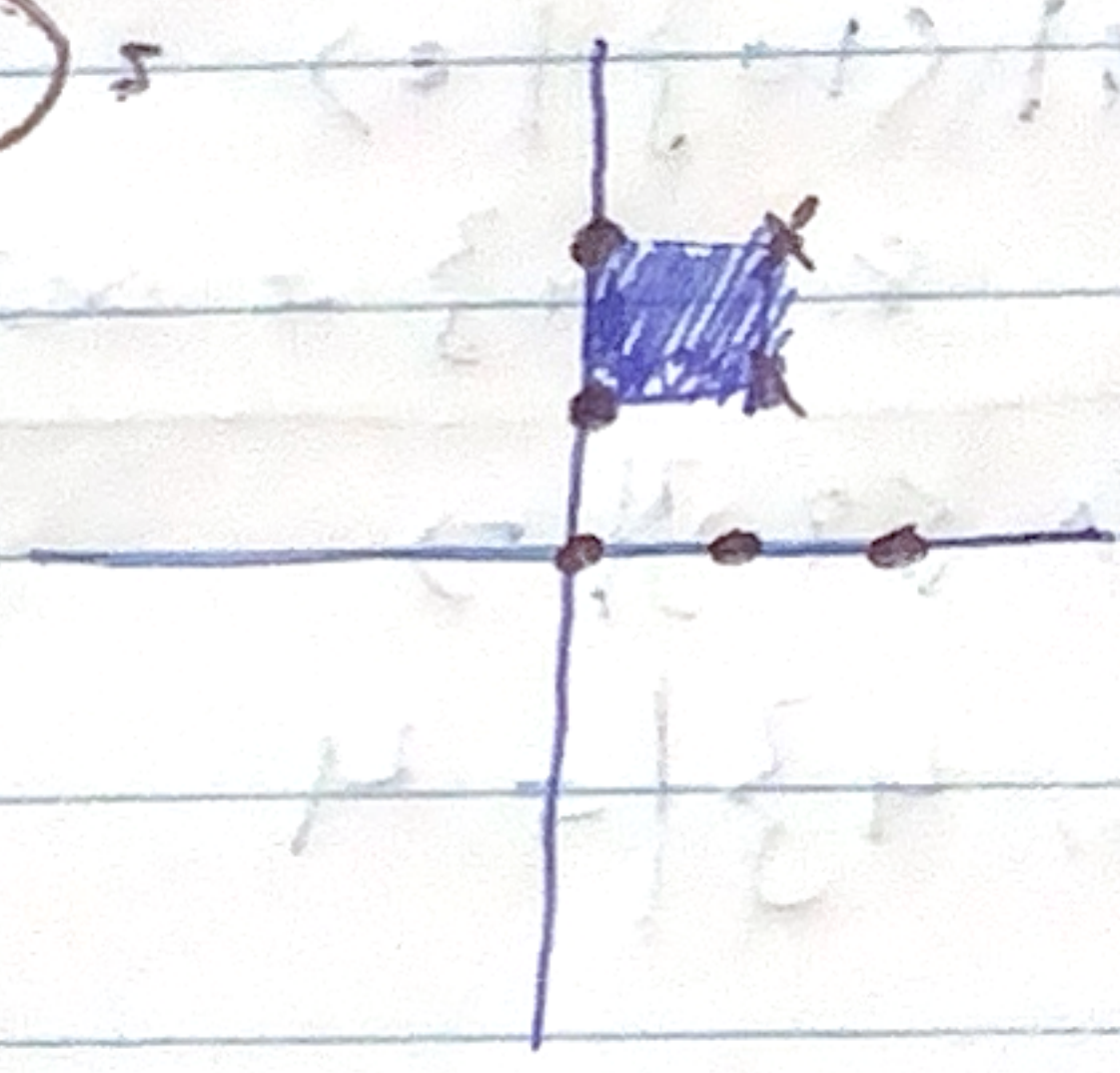
(3)



(6) because $|\emptyset| = 0 \Rightarrow |A| \times |\emptyset| = 0$

$\frac{3}{4}$ since only $|\emptyset| = 0$, thus it must be the case that $A \times \emptyset = \emptyset$

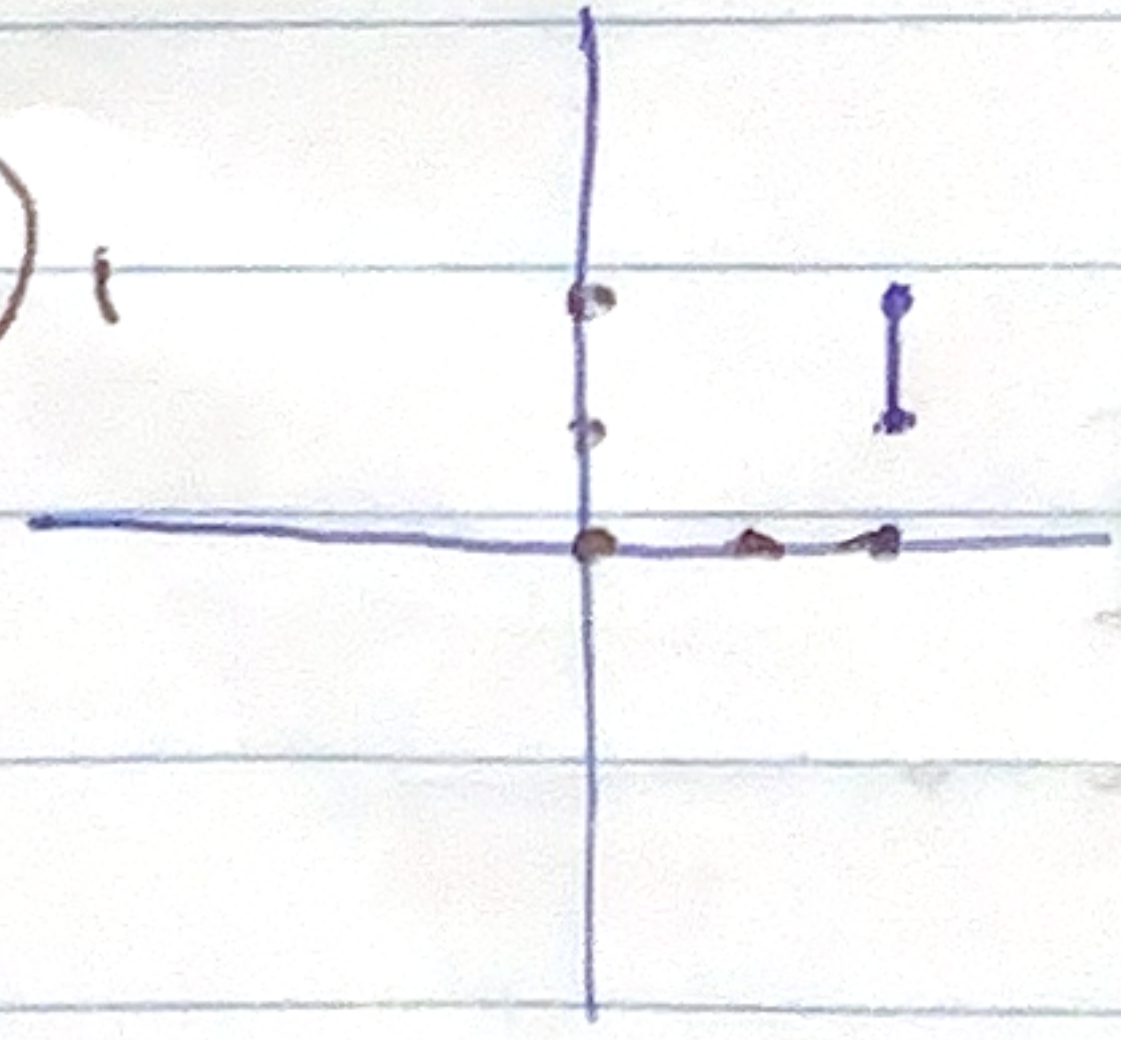
(7, 1)



$$[0,1] \times [1,2] \Rightarrow [(0,1), (0,2), (1,1), (1,2)]$$

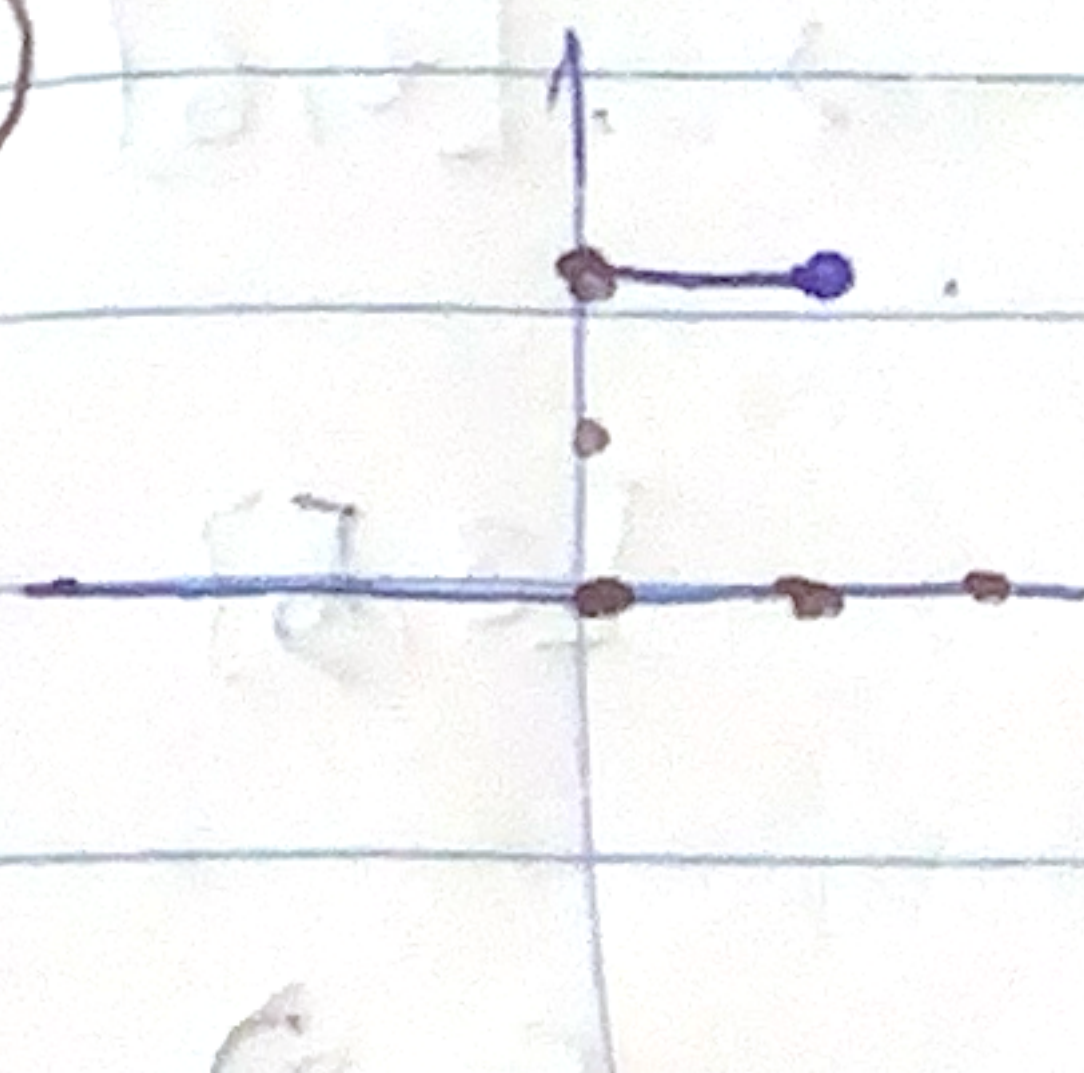
$$[(0,1), (0,1,1), (0,2), (1,1), (1,1,1), (1,1,2), (1,2)]$$

(7, 2)



$$C = \{e\} \times [1,2] \Rightarrow [(e,1), (e,1,1), (e,1,2), (e,2)]$$

7.3



$$[0,1] \times \{2\} \Rightarrow$$

$$[(0,2), (0.1,2), (0.3,2), \dots, (1,2)]$$

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$$a \rightarrow T$$

$$e \rightarrow F$$

$$i \rightarrow F$$

$$b \rightarrow F$$

$$f \rightarrow T$$

$$j \rightarrow T$$

$$c \rightarrow F$$

$$g \rightarrow T$$

$$d \rightarrow F$$

$$h \rightarrow T$$

9 $\frac{1}{2}$ $a \rightarrow \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

$$b \rightarrow [\emptyset] \quad b \rightarrow \{\emptyset\}$$

$$c \rightarrow [\emptyset, \{\emptyset, \{\emptyset\}\}]$$

$\frac{2}{3}$ $d \rightarrow \{\emptyset, \underbrace{\{\{0,1\}\}}, \underbrace{\{\{0,1\}\}}, \underbrace{\{\{0,1\}, \{0,1\}\}}\}$

no duplicate elements in sets

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Everything that we can imagine in 2D plane

is $P(R^2)$

included in

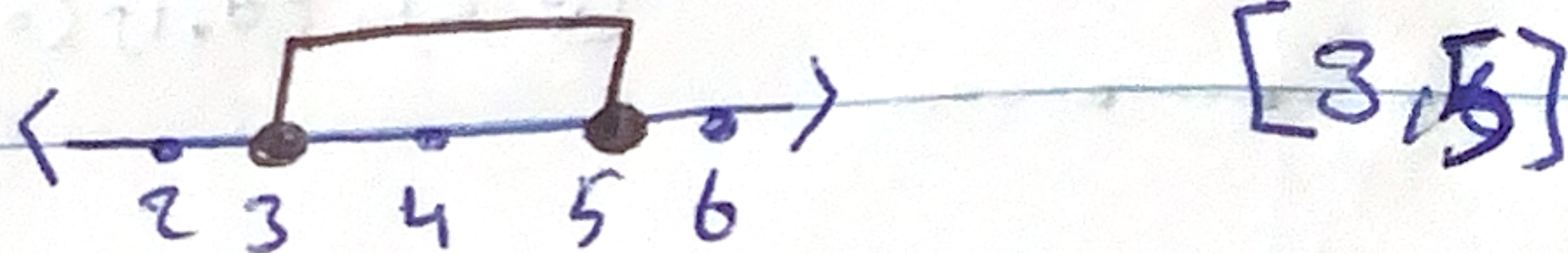
and everything we can't imagine.

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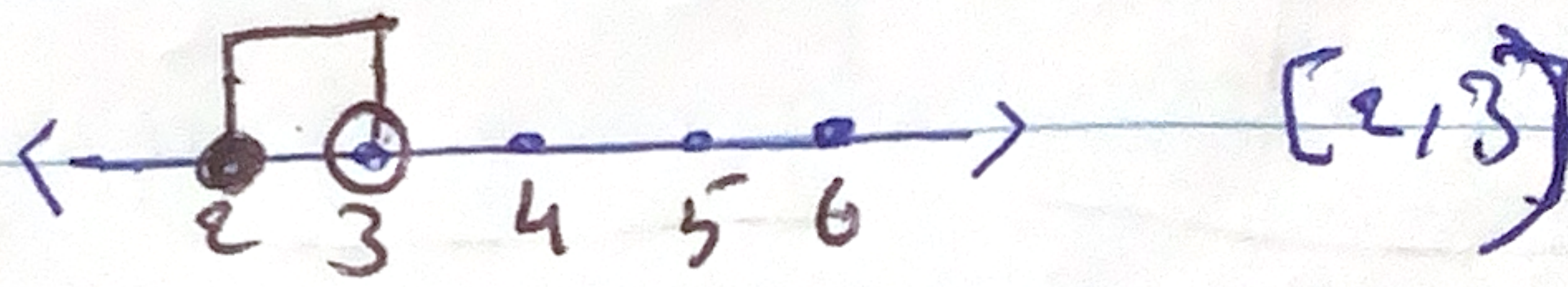
a)



b)



c)



d)

